

Start Here - Satellite1 Ultra

DO NOT PRINT THE FULL ENCLOSURE YET.

PRINT AND COMPLETE THE CALIBRATION PARTS FIRST.

Satellite1 Ultra is a serviceable passive-radiator enclosure for the FutureProofHomes Satellite1 **Batch 1** development kit: Core rev4.1 and HAT rev4.1 / R2024.12.06. Satellite1.1 / Batch 2 is not supported.

Status: DIGITAL_PROTOTYPE_READY only after every digital gate passes. No physical specimen has been validated. Fit, sealing, acoustics, thermals, Wi-Fi, microphones, buttons, LEDs, and wake-word performance are REQUIRES_PHYSICAL_VALIDATION.

You need

- An enclosed 256 x 256 x 256 mm or larger FDM printer capable of ASA.
- 0.4 mm nozzle, dry ASA, TPU 95A, and a documented PETG alternative.
- Digital calipers (0.01 mm display), 0.01 g scale, 2.0 mm hex driver, M3 insert tip, wire tools, ESD protection, and basic acoustic test equipment.
- Everything in BOM.csv, FASTENERS.csv, and GASKETS.csv.

Exact order

1. Read START_HERE_CALIBRATION_GUIDE.pdf.
2. Print seven coupon files plus cable_gland.3mf.
3. Measure, edit CALIBRATION_INPUT_TEMPLATE.yaml, and run `make calibrated-release`.
4. Reprint affected coupons and pass every calibration check.
5. Follow PRINTING_GUIDE.pdf, then ASSEMBLY_GUIDE.pdf.
6. Complete TESTING_AND_COMMISSIONING_GUIDE.pdf before normal use.

The project is advanced: expected builder difficulty is 4/5. Allow several days for printing plus calibration and test time.

Exploded parts identification

Removal directions match the service sequence; official mechanics are shown in grey.



FRONT = -Y | TOP = +Z | units: mm

Exploded Satellite1 Ultra

Source commit 45cfb9dd3357. No physical validation has been performed.